

Deep learning avec Pytorch

Partie BONUS : Gérer un projet IA

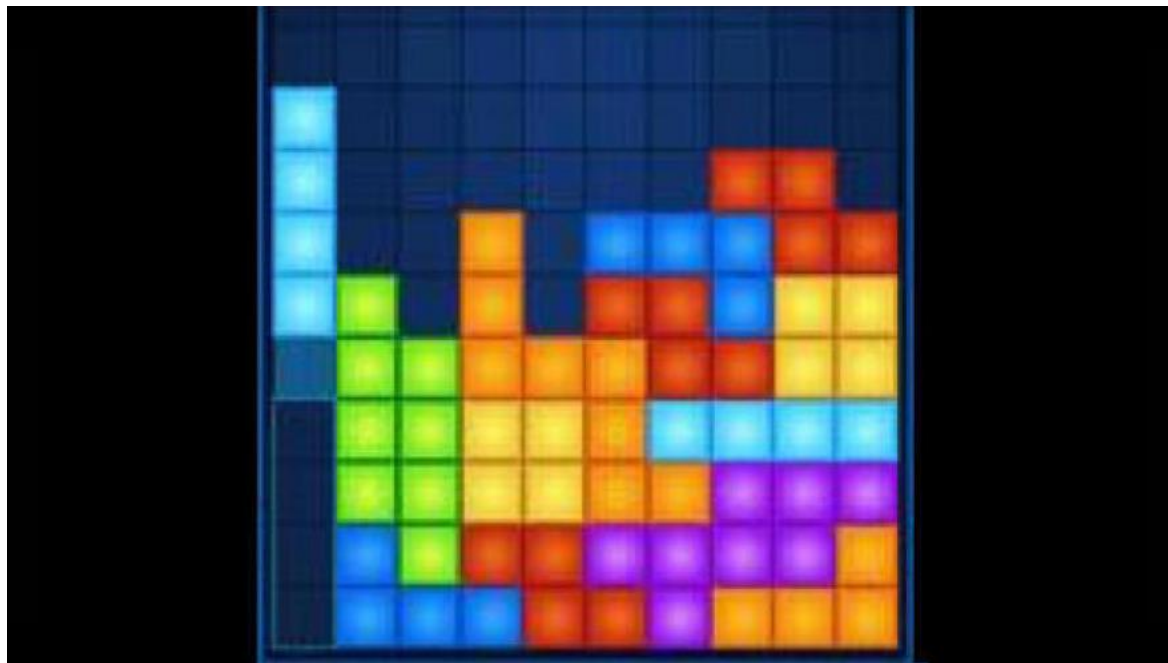


Présenté par **Morgan Gautherot**



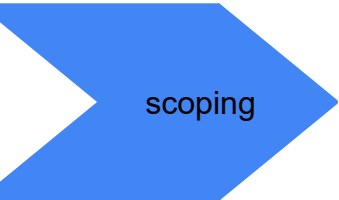
Exemple

Utiliser l'IA pour battre le champion du monde





Scoping



scoping

Objectif:

- Obtenir plus de 8 millions de points
- Détruire plus de 2 000 lignes



La vie d'un projet ML

scoping

Inventaire
des données

Pas de données



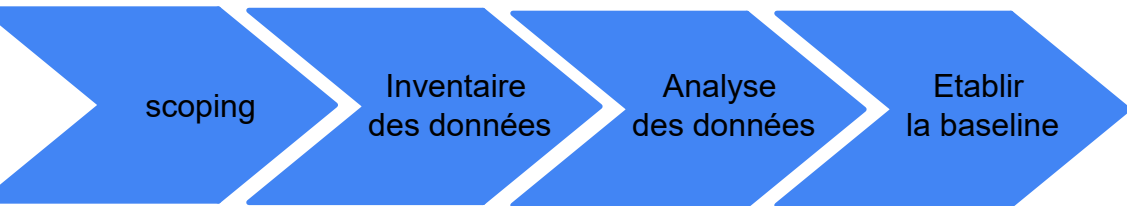
La vie d'un projet ML



Pas de données



La vie d'un projet ML



Création d'un système expert



Déterminer chaque position

```
File Edit Selection View Go Run Terminal Help
tetris_godmod

EXPLORER
  TETRIS_GODMOD
    __pycache__
    > pytest_cache
    > artefacts
    > myenv
    > scraping
    > src
    > __pycache__
    bot.py M
    config.py
    game.py
    save_examples.py
    score.py
    tetrominoes.py
    windows.py
    > tests
    > __pycache__
    > data
    test_bot.py
    test_game.py
    test_score.py
    test_scraping.py
    test_tetrominoes.py
    > train
    training_model_ten_lines.ipynb
    > .gitignore
    main_bot.py M
    main_seed.py
    main.py
    README.md
    requirements.txt
    scenario.py
    test.py
    y_0_dataframe.csv U

  > OUTLINE
  > TIMELINE

main_bot.py M bot.py M X
src > bot.py > ExpertBot > play

230 new_line = pd.DataFrame([[width,
231                             rotation,
232                             position_cost]],
233                             columns=['id_column',
234                                     'id_rotation',
235                                     'cost'])
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return move_cost_dataframe

def play(self)->None:
    """
    place the tetrominoes minimizing the hole
    """
    move_cost_dataframe = self.move_estimation(self.tetris.current_tetromino,
                                              self.tetris.game_board_matrix,
                                              recursivity_id=False)

    minimum_move_cost_dataframe = move_cost_dataframe[move_cost_dataframe['cost']==move_cost_dataframe['cost'].min()]
    #print(minimum_move_cost_dataframe)
    next_position = minimum_move_cost_dataframe.sample(1)
    #print(next_position)

    new_shape = self.tetris.current_tetromino.rotate(self.tetris.current_tetromino.tetromino_shape,
                                                    times=next_position.loc[0, 'id_rotation'])
    self.tetris.current_tetromino.tetromino_shape = new_shape

    self.tetris.current_tetromino.tetromino_position = (0, next_position.loc[0, 'id_column'])

    self.tetris.hard_drop()

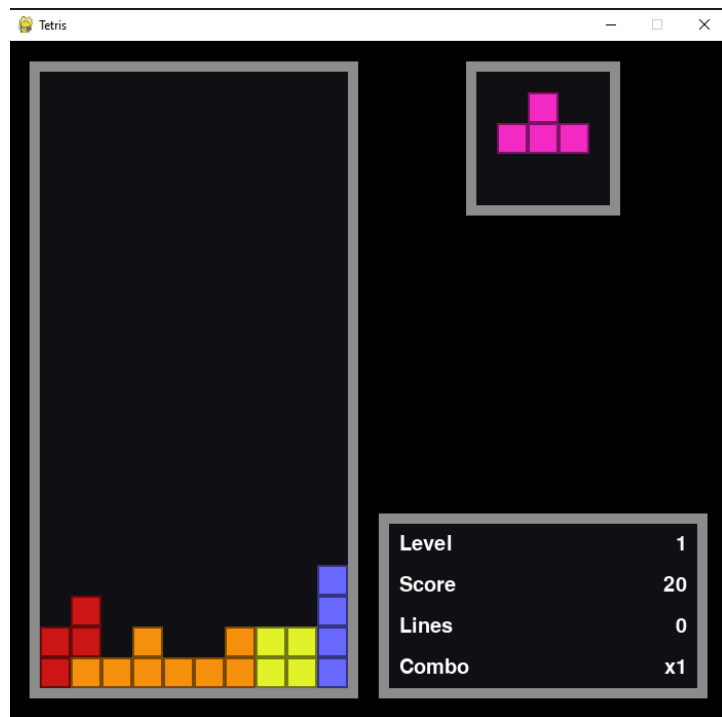
    #time.sleep(0.5)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
position_cost = self.compute_cost_with_position(current_tetromino_copy,
File "C:\Users\Gautherot\Documents\code\code_videos\tetris_godmod\tetris_godmod\src\bot.py", line 157, in compute_cost_with_position
self.tetris.tetris_window.redraw(matrix_and_tetromino,
File "C:\Users\Gautherot\Documents\code\code_videos\tetris_godmod\tetris_godmod\src\windows.py", line 62, in redraw
self.blit_info()
File "C:\Users\Gautherot\Documents\code\code_videos\tetris_godmod\tetris_godmod\src\windows.py", line 226, in blit_info
area.fill(
KeyboardInterrupt
(myenv) PS C:\Users\Gautherot\Documents\code\code_videos\tetris_godmod\tetris_godmod> python .\main_bot.py]
```

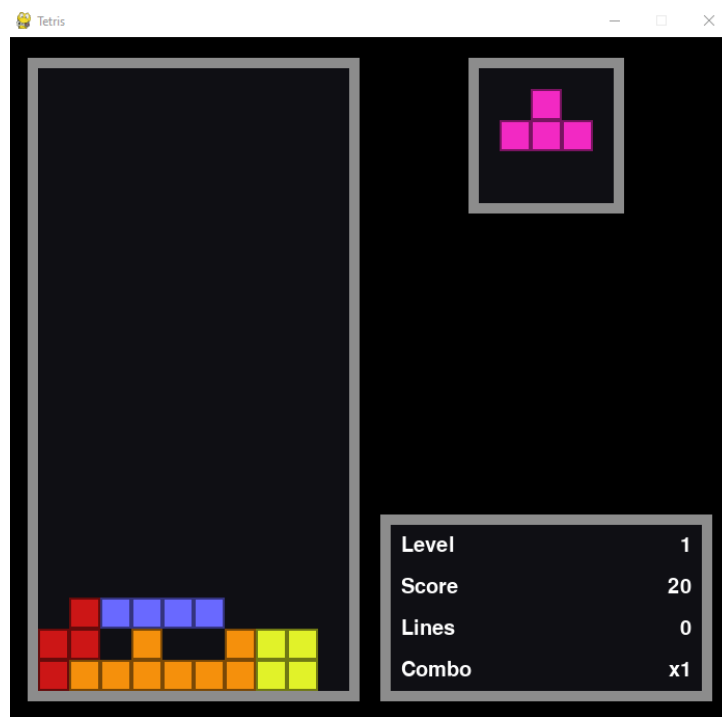


Créer une règle

Bien placé

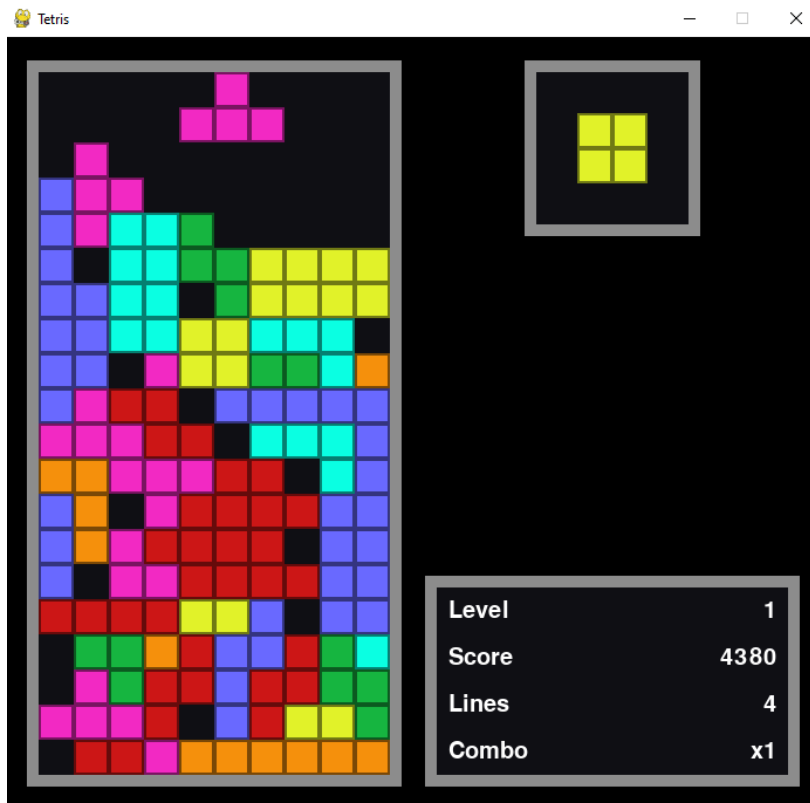


Mal placé





Ne pas laisser de trous



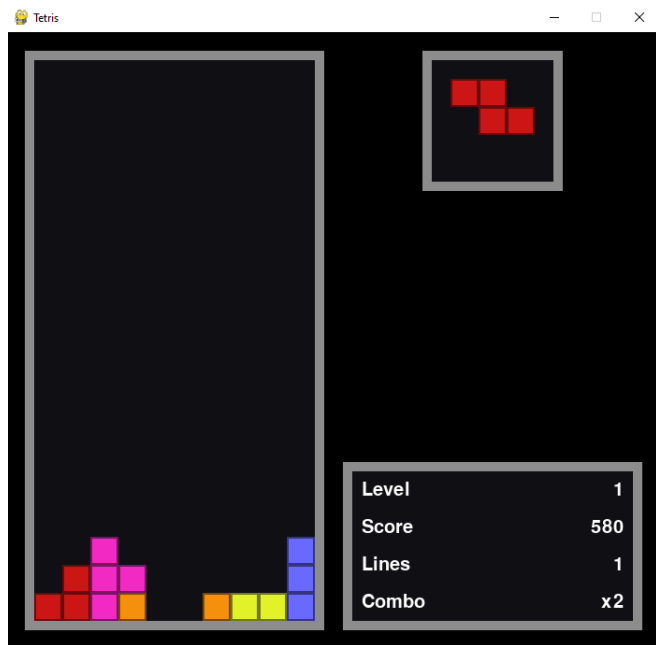


Fonction de coût

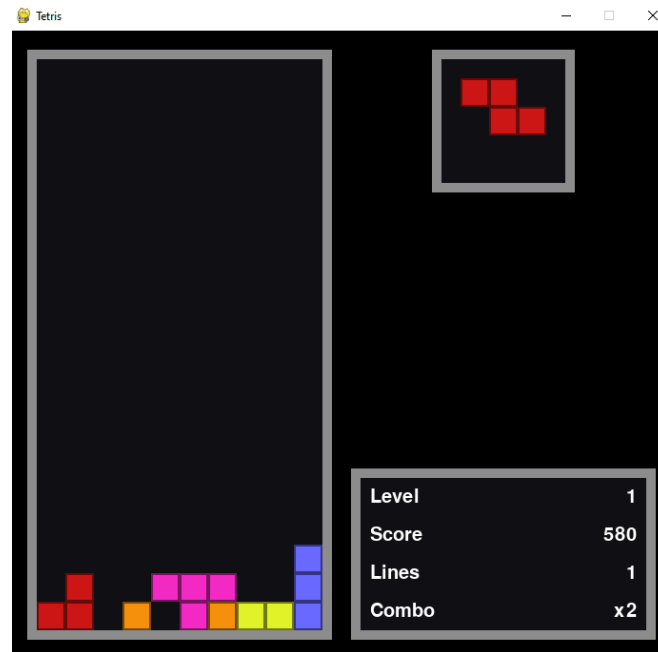
- Erreur = nombre de trous



Calculer l'erreur



Erreur = 0



Erreur = 1



Baseline en action

```
9
10
11 tetris = Tetris(take_picture = False,
12               training_id=3,
13               display=True,
14               seed=1132)
15
16 #bot = RandomBot(tetris)
17 bot = ExpertBot(tetris)
18
19 #bot = DeepBot(tetris,
20               # display=True)
21
22 while not(tetris.game_over):
23
24     # Time of the game
25     #tetris.tetromino_falls_over_time()
26     matrix_and_tetromino = tetris.add_tetromino_to_game_board_matrix(tetris.current_tetromino,
27                                                                     tetris.game_board_matrix)
28     tetris.tetris_window.redraw(matrix_and_tetromino,
29                               tetris.next_tetromino)
30     #random_bot(tetris)
31     tetris.user_action()
32
33 bot.play()
34 #print(count_hole_number(tetris.matrix))
35 #custom_metric(tetris.matrix)
36 #system_expert(tetris)
37
38 print(tetris.tetris_score.score)
39 print(tetris.tetris_score.lines)
40
41
42
43 while True:
44     a = 1
45
46
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(myenv) PS C:\Users\Vaatherot\Documents\code\code_videos\tetris_godmod\tetris_godmod> python .\main_bot.py
pygame 2.5.2 (SDL 2.28.3, Python 3.11.6)
Hello from the pygame community. https://www.pygame.org/contribute.html
Traceback (most recent call last):
  File "C:\Users\Vaatherot\Documents\code\code_videos\tetris_godmod\tetris_godmod\main_bot.py", line 33, in <module>
    bot.play()
  File "C:\Users\Vaatherot\Documents\code\code_videos\tetris_godmod\tetris_godmod\src\bot.py", line 266, in play
    time.sleep(0.5)
KeyboardInterrupt
(myenv) PS C:\Users\Vaatherot\Documents\code\code_videos\tetris_godmod\tetris_godmod>
```

Ln 44, Col 10 Spaces:4 UTF-8 CR LF Python 3.11.7 (64-bit) 1364 28/01/2024



La vie d'un projet ML

scoping

Inventaire
des données

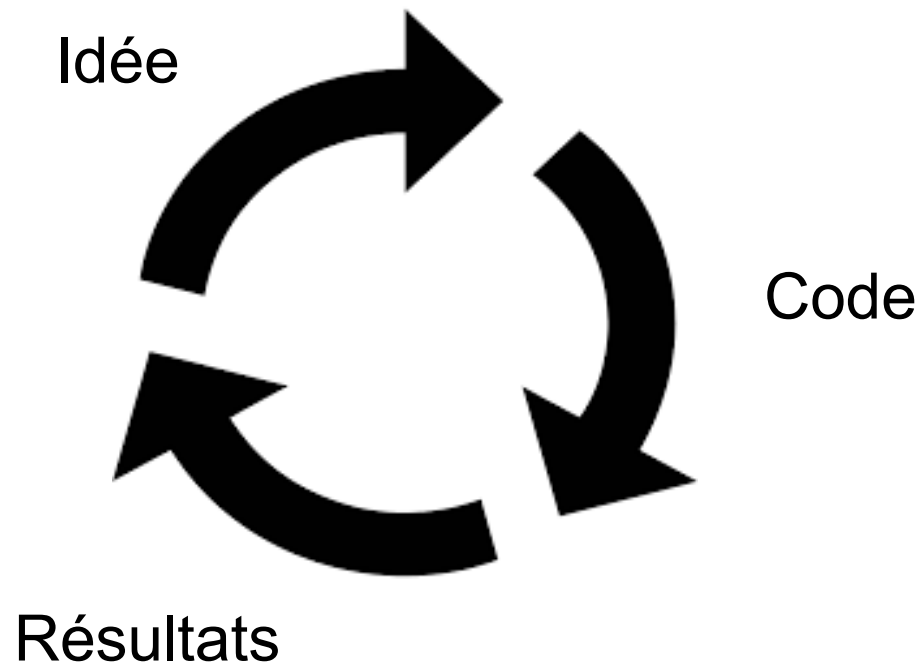
Analyse
des données

Etablir
la baseline

Itérer sur
la complexité

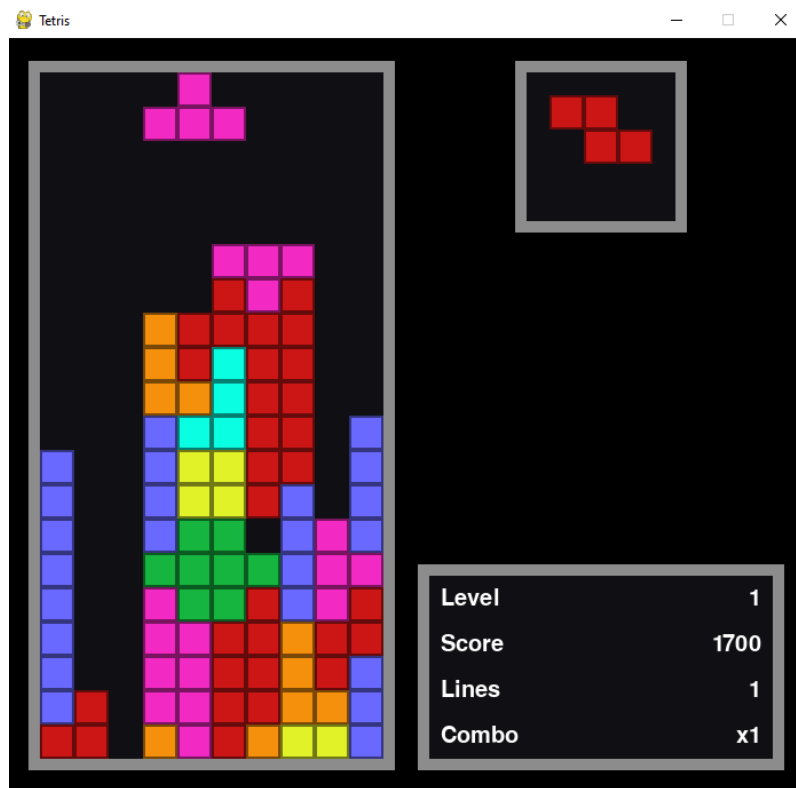


Itérations





Idée 1





Fonction de coût

- Erreur = nb_trous
+ hauteur_max
– hauteur_min



Résultats

```
def custom_metric(self, game_matrix:dict)->float:
    """
    Mix different cost function
    """
    nb_hole = self.count_hole_number(game_matrix)
    max_height = self.count_max_height(game_matrix)
    min_height = self.count_min_height(game_matrix)
    line = self.is_line(game_matrix)
    hole_colmun = self.count_hole_column(game_matrix)

    return nb_hole + max_height - min_height

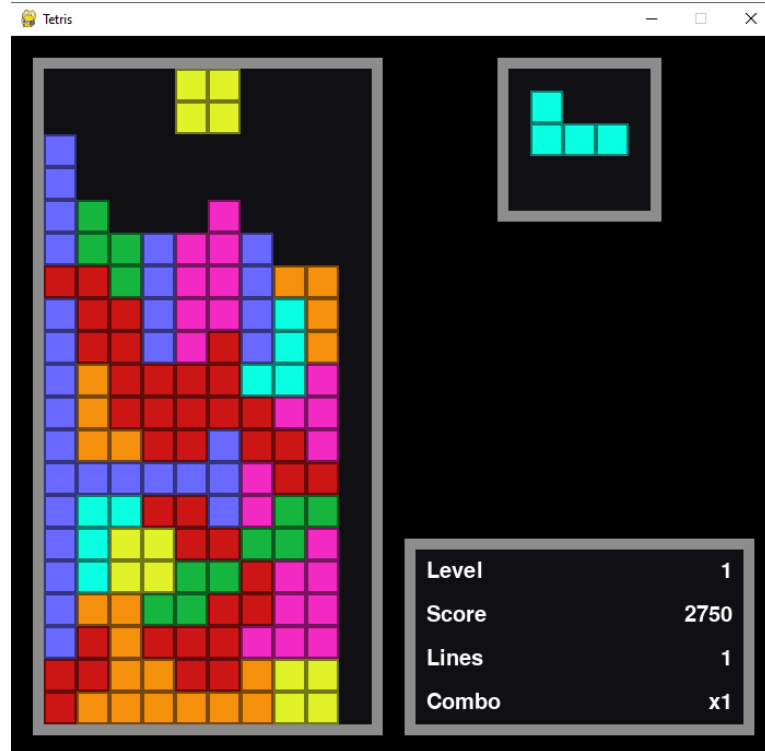
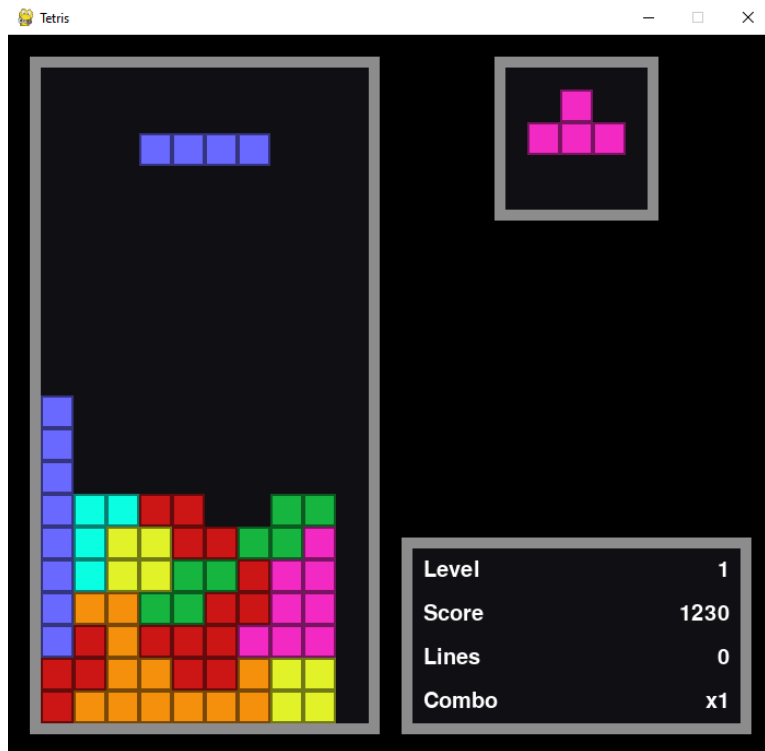
def count_hole_number(self, game_matrix:dict)->float:
    """
    Count the number of hole in the tetris matrix.
    A hole is an empty space cover by a tetrominoes
    """
    count_hole = 0
    number_hole = 0
    for width in range(config.MATRIX_WIDTH):
        number_hole = 0
        for height in range(config.MATRIX_HEIGHT-1, -1, -1):
            if game_matrix[height, width] is None:
                number_hole +=1
            if not(game_matrix[height, width] is None):
                count_hole += number_hole
```

File "C:\Users\Gautherot\Documents\code_videos\tetris_godmod\tetris_godmod\src\windows.py", line 85, in draw_surface
tetromino_block = self.block(board_game[(y, x)])
.....
File "C:\Users\Gautherot\Documents\code_videos\tetris_godmod\tetris_godmod\src\windows.py", line 118, in block
list(
File "C:\Users\Gautherot\Documents\code_videos\tetris_godmod\tetris_godmod\src\windows.py", line 120, in <lambda>
lambda c: min(255, int(c * 1)),
.....
KeyboardInterrupt
(myenv) PS C:\Users\Gautherot\Documents\code_videos\tetris_godmod\tetris_godmod> []

Ln 118, Col 49 Spaces: 4 UTF-8 LF Python 3.8.17 (RL) conda 12:51 28/01/2024

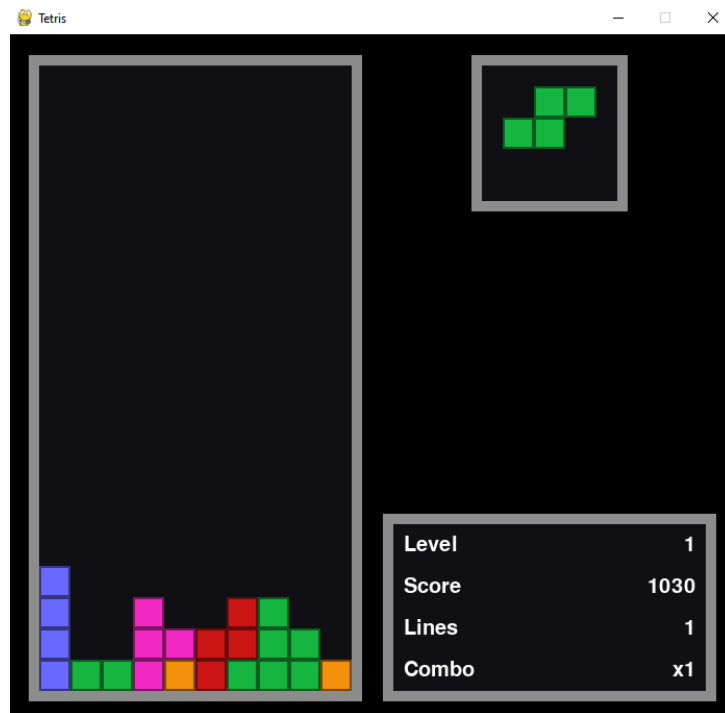
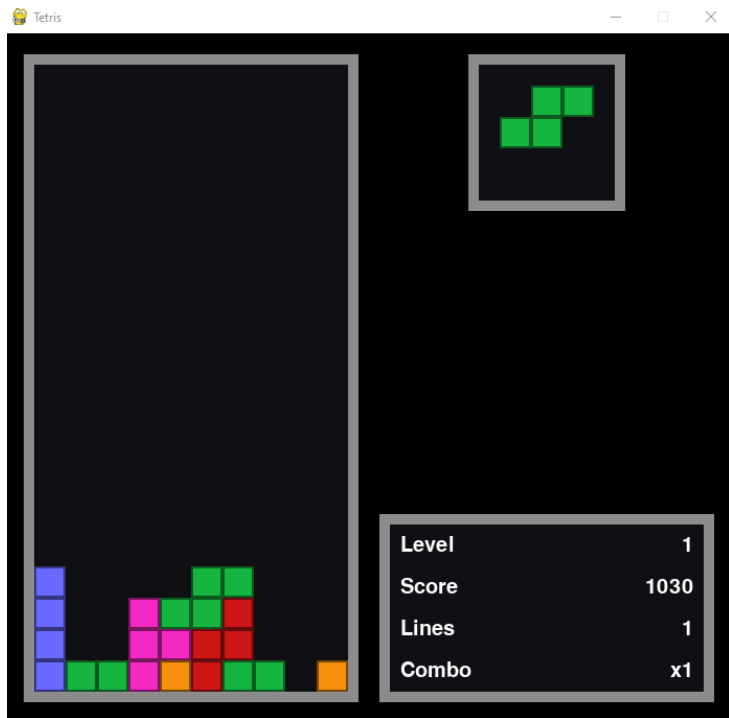


Idée 2



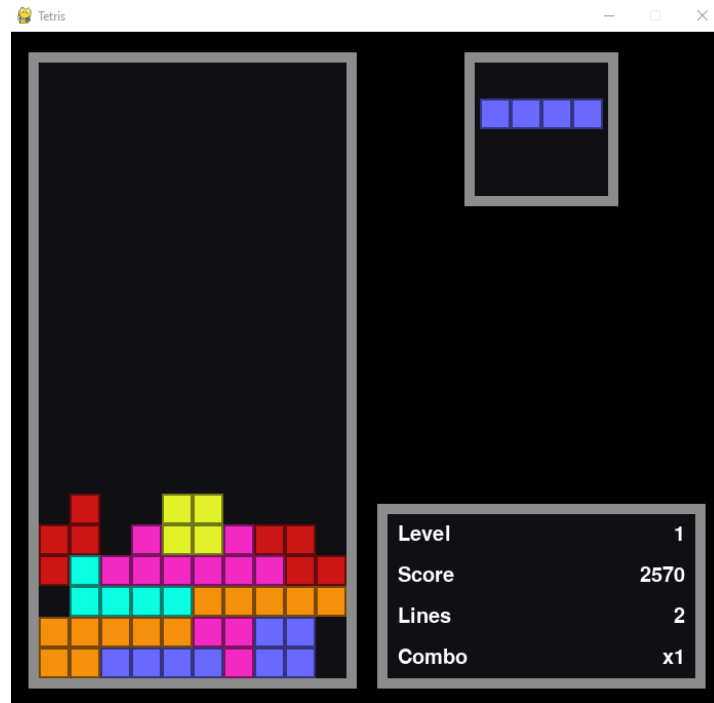
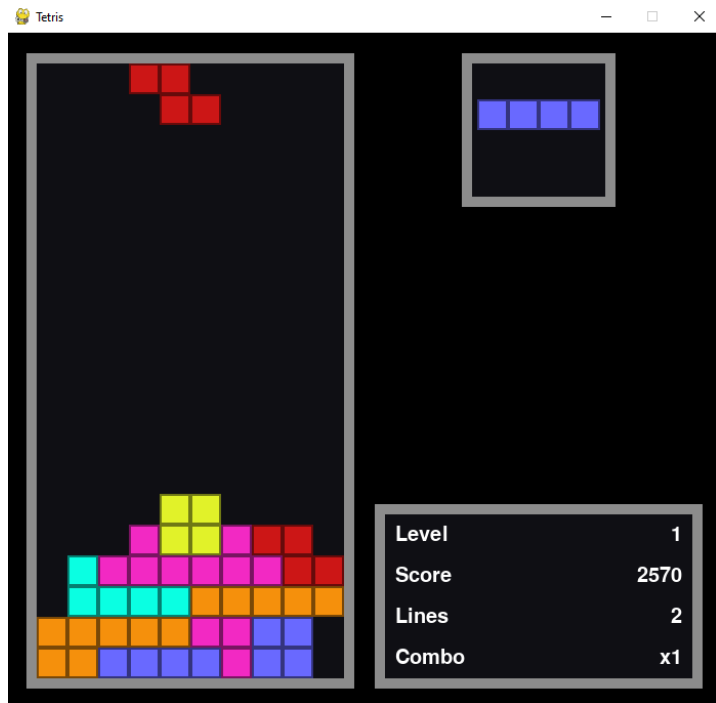


Idée 3



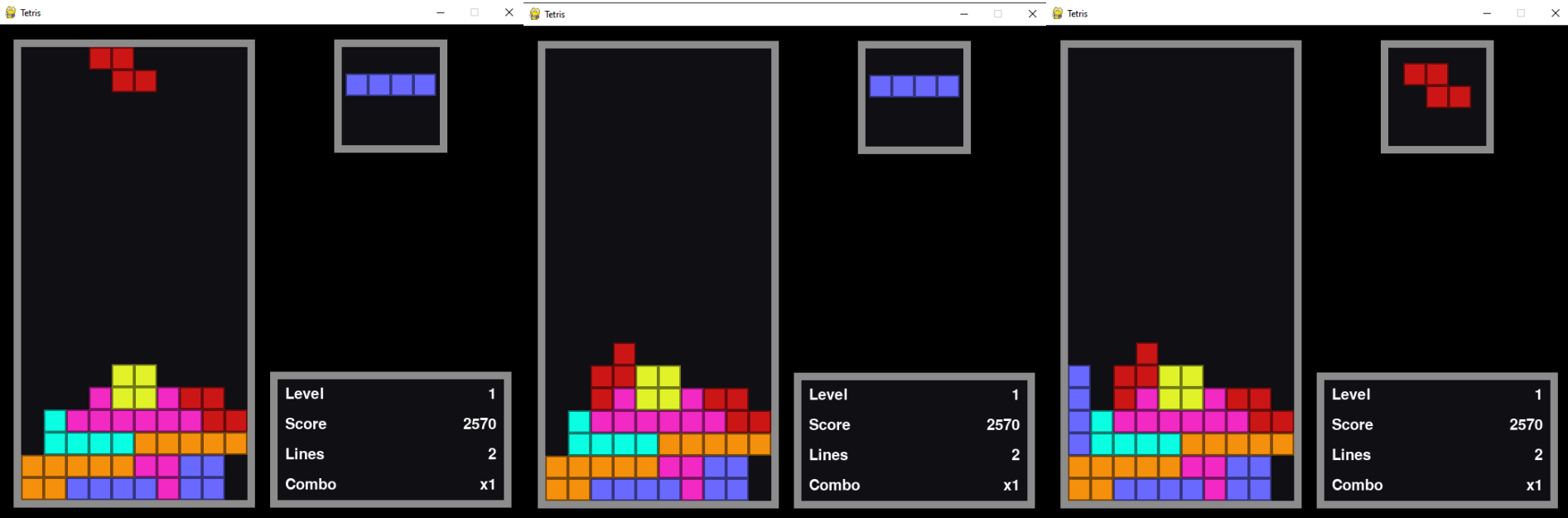


Idée 4





Idée 4





Double visualisation

The image shows a VS Code editor window with a Python file named `main_bot.py` open. The file is part of a project named `TETRIS_GODMOD`. The code in `main_bot.py` is as follows:

```
231 rotation,
232 position_cost]],
233 columns=['id_column',
234          'id_rotation',
235          'cost'])
236
237
238
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240
241 return move_cost_dataframe
242
243 def play(self)->None:
244     """
245     place the tetrominoes minimizing the hole
246     """
247
248     move_cost_dataframe = self.move_estimation(self.tetris.current_tetromino,
249                                               self.tetris.game_board_matrix,
250                                               recursivity_id=True)
251
252     minimum_move_cost_dataframe = move_cost_dataframe[move_cost_dataframe['cost']--move_cost_dataframe['cost'].min()]
253     #print(minimum_move_cost_dataframe)
254     next_position = minimum_move_cost_dataframe.sample(1)
255     #print(next_position)
256
257     new_shape = self.tetris.current_tetromino.rotate(self.tetris.current_tetromino.tetromino_shape,
258                                                       times=next_position.loc[0, 'id_rotation'])
259     self.tetris.current_tetromino.tetromino_shape = new_shape
260
261     self.tetris.current_tetromino.tetromino_position = (0, next_position.loc[0, 'id_column'])
262
263     self.tetris.hard_drop()
264
265     time.sleep(0.5)
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```

The terminal output shows the following:

```
(myenv) PS C:\Users\Gautherot\Documents\code\code_videos\Tetris_godmod\tetris_godmod> python .\main_bot.py
pygame 2.5.2 (SDL 2.28.3, Python 3.11.6)
Hello from the pygame community. https://www.pygame.org/contribute.html
2570
2
Traceback (most recent call last):
  File "C:\Users\Gautherot\Documents\code\code_videos\Tetris_godmod\tetris_godmod\main_bot.py", line 43, in <module>
    while True:
KeyboardInterrupt
(myenv) PS C:\Users\Gautherot\Documents\code\code_videos\Tetris_godmod\tetris_godmod>
```

The status bar at the bottom indicates the file is at line 250, column 72, with 4 spaces, UTF-8 encoding, LF line endings, Python 3.11.7 (RL) interpreter, and a timestamp of 14:58 on 28/01/2024.



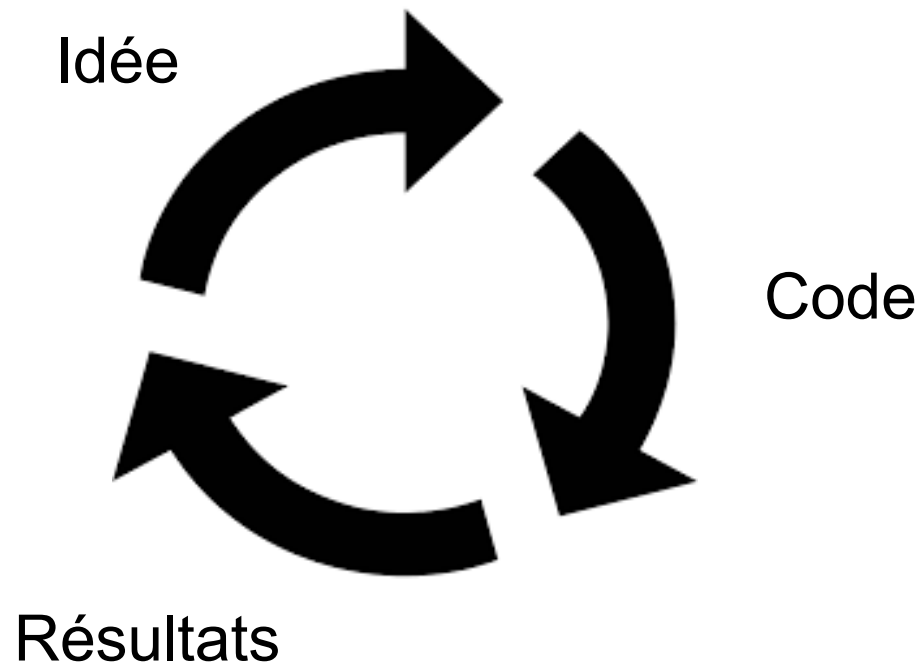
Résultats

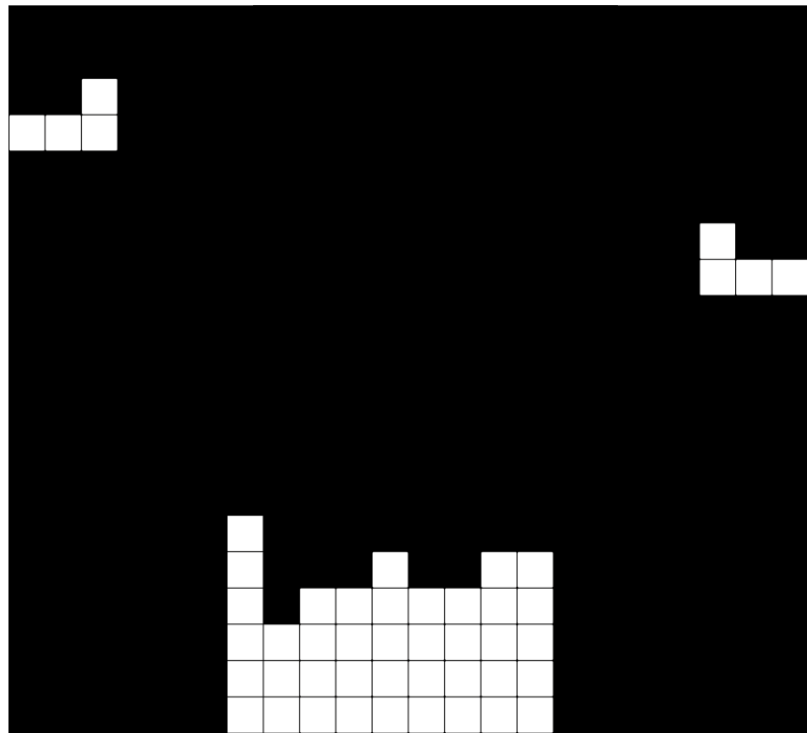
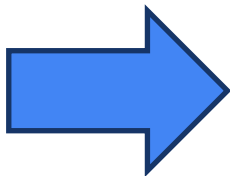
Score: 1 447 207

Nombre de lignes : 812



Itérations

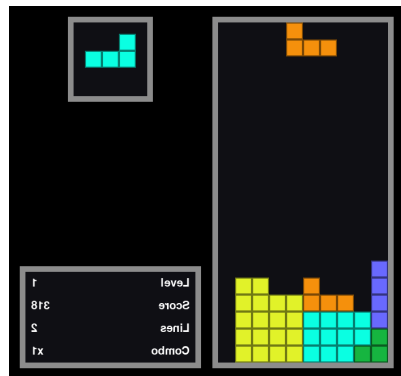




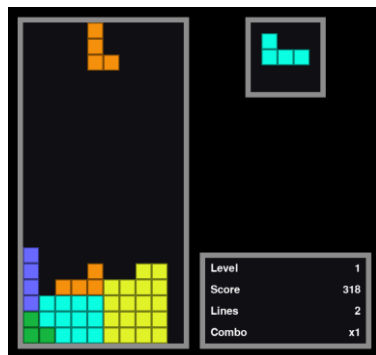


Enregistrement des données de sorties

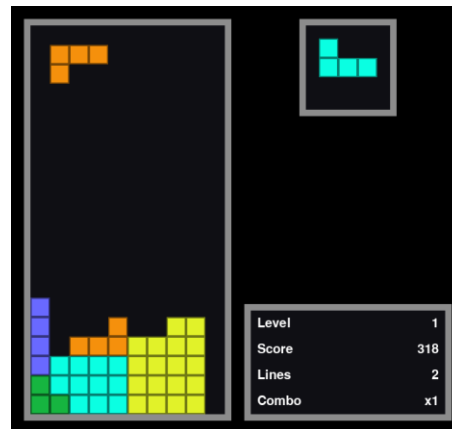
Etat initial



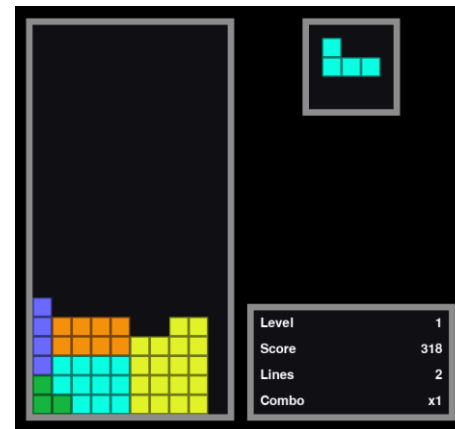
Rotation = 2



Colonne = 1



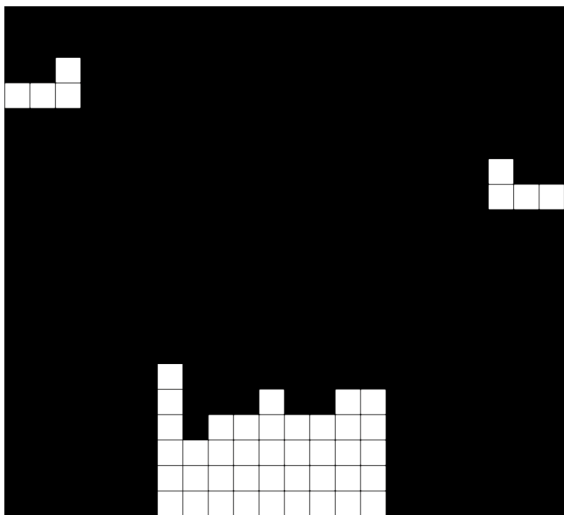
Etat final



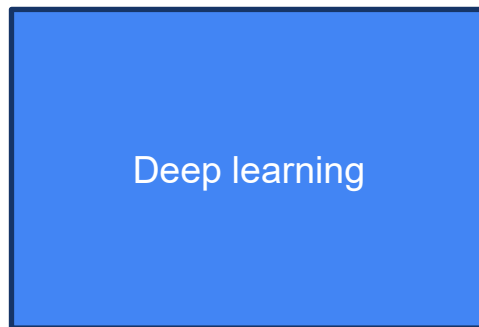


Entraînement d'un modèle

Données d'entrées



Modèle



Données de sortie

Rotation = 2

Colonne = 1



Quantité des données

Nombre images

Performances

Temps

5 000

Nulle

2 heures

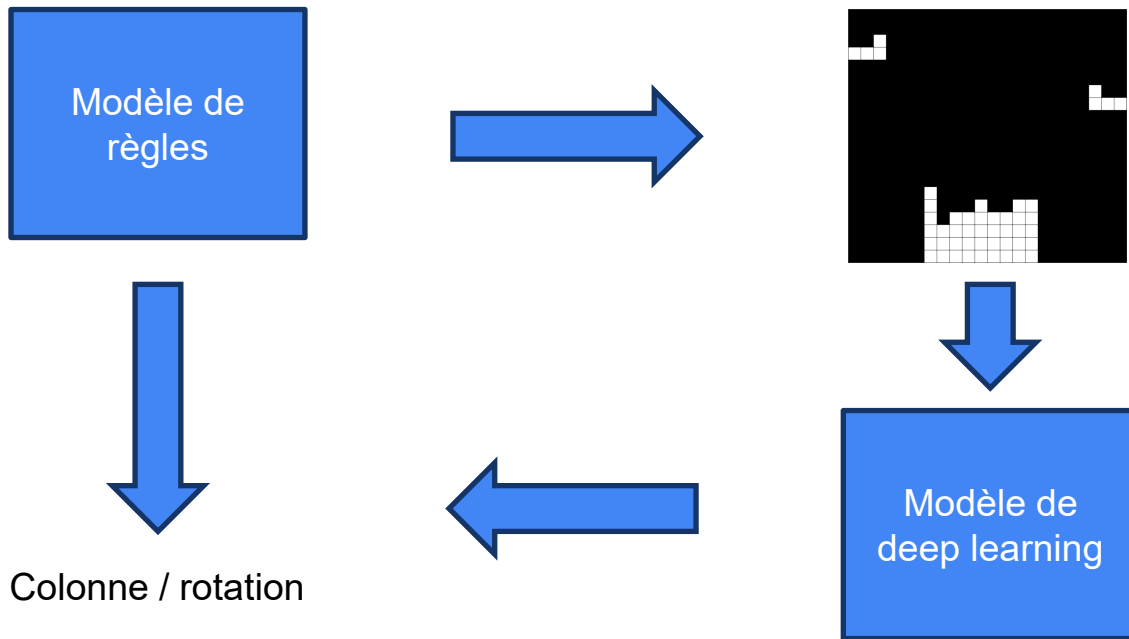
100 000

Très faible

2 jours



Génération de données





Quantité des données

Nombre images

Performances

Temps

10 000

Nulle

5 minutes

100 000

Très faible

1 heure

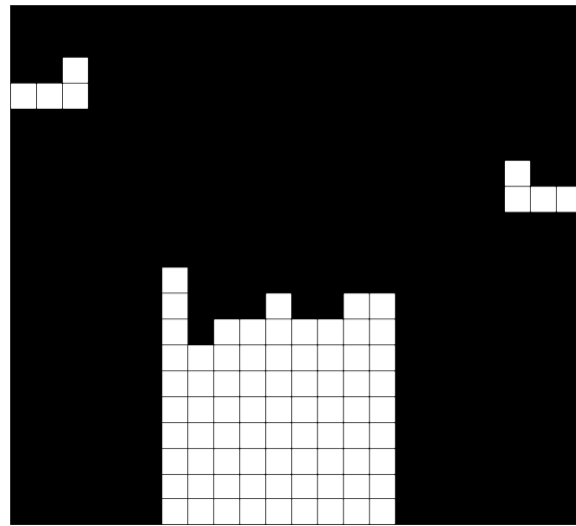
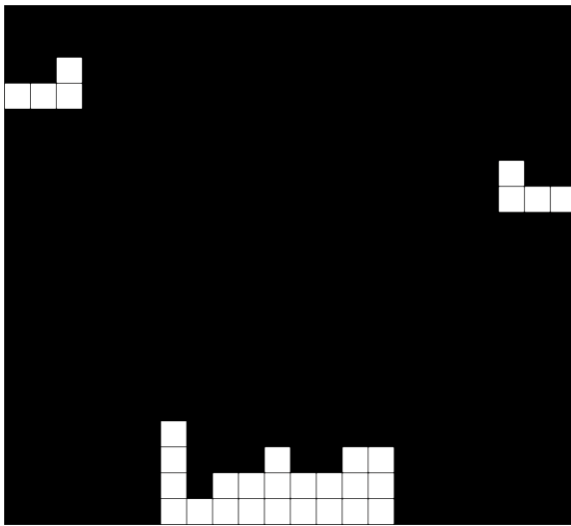
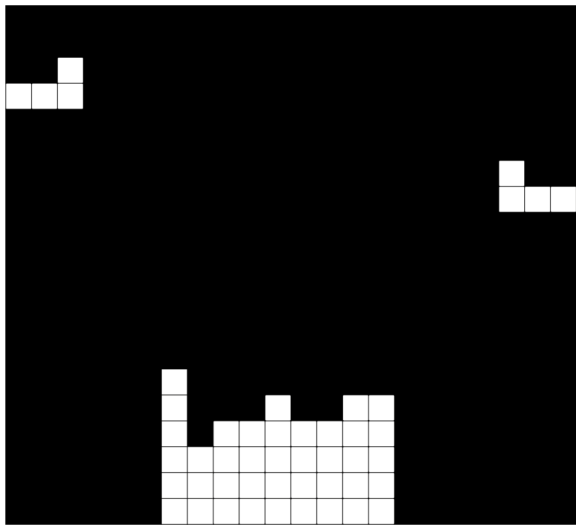
2 000 000

Faible

1 journée



Data augmentation





Quantité des données

	Nombre images	Performances	Temps
	10 000	Nulle	5 minutes
	100 000	Très faible	1 heure
	2 000 000	Faible	1 journée
Data augmentation	30 000 000	Moyenne	



Résultats

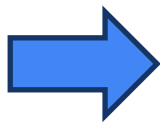
Score: 750 811

Nombre de lignes : 580

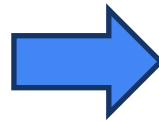


Qualité des données

Données



Modèle de
machine learning

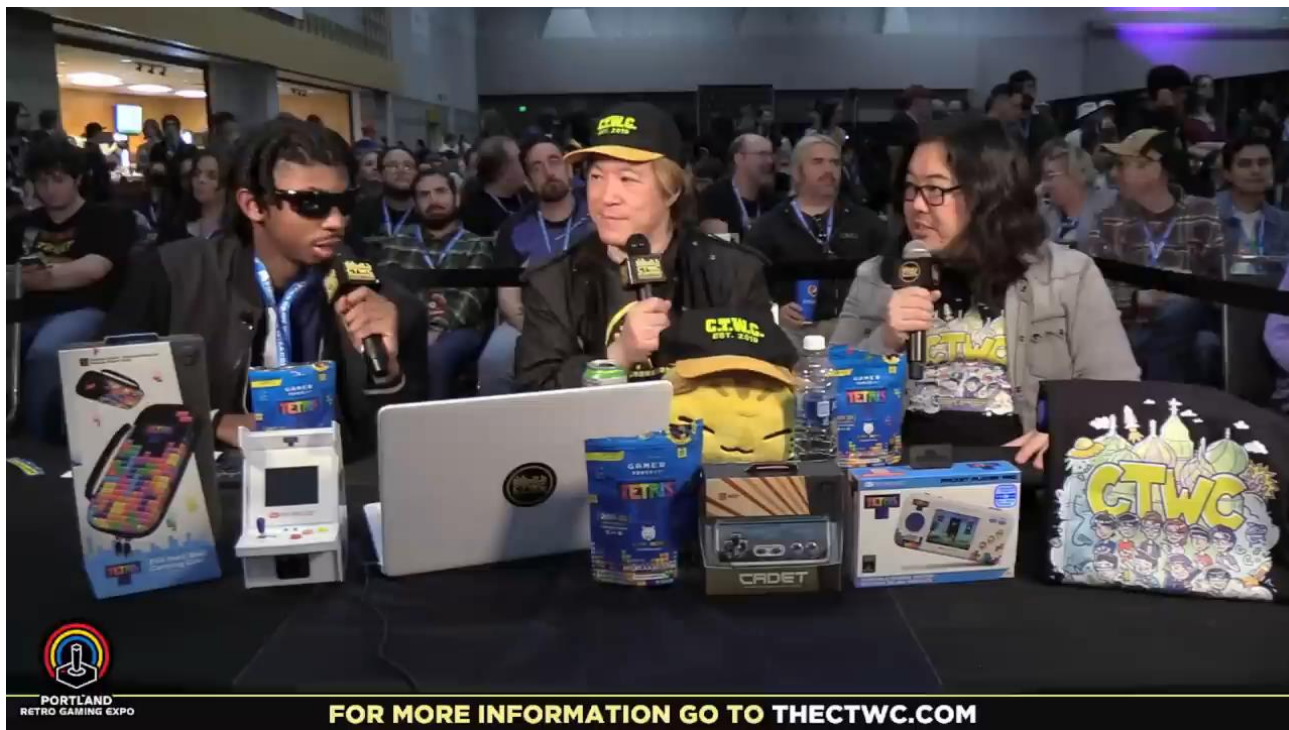


Prédiction





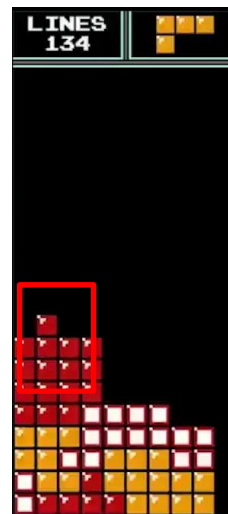
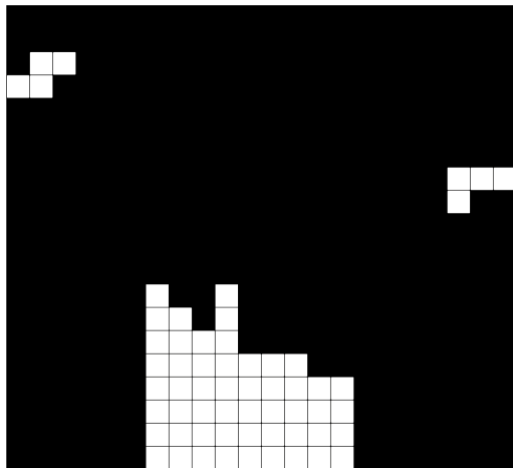
Parties de professionnelles





Création du dataset

Données d'entrée



Rotation = 1

Colonne = 1



Fine tuning

Modèle aléatoire

Performance nulle

Entraînement



Modèle entraîné

Performance moyenne

Fine tuning

Modèle
pré-entraîné

Performance moyenne

Entraînement



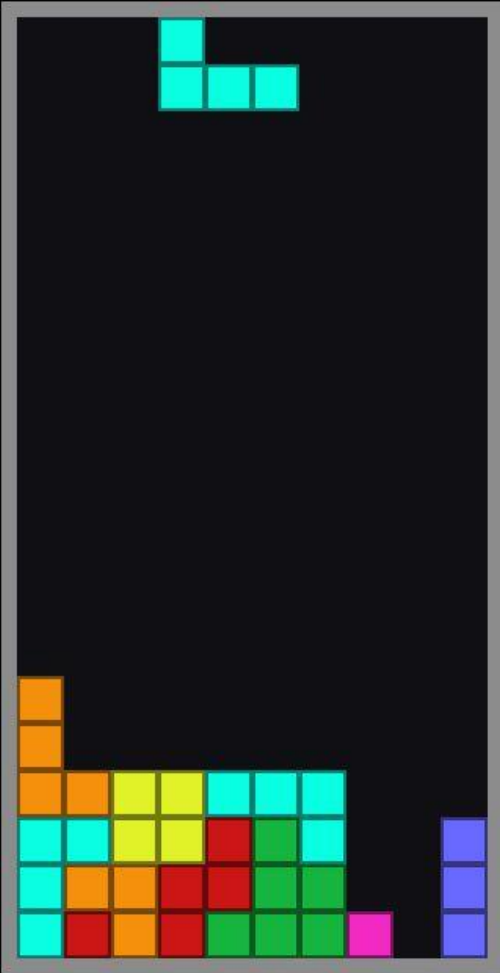
Modèle entraîné

Performance incroyable



Résultats

Tetris



Level 202
Score 12926820
Lines 2011
Combo x2